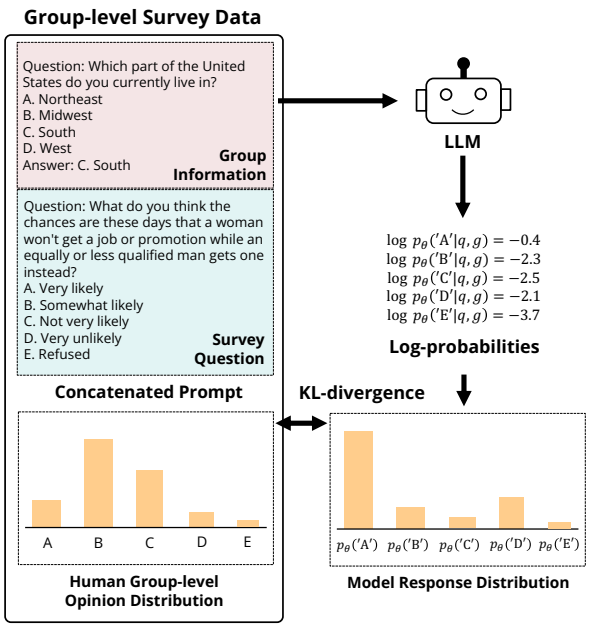

Figure 2: Proposed supervised fine-tuning setup with a survey response dataset such as SubPOP. Survey data is 3-tuple of a survey question, target subpopulation information, and the observed human opinion distribution (*i.e.* how subjects in the group responded to the given question). The training objective, $\mathcal{L}(\theta)$, is a forward KL divergence loss on language model predicted distribution of question option likelihoods; our loss guides the model predictions to match the response distribution of the specified human subpopulation.

available responses are: *A. Very likely, B. Somewhat likely, C. Not very likely, D. Very unlikely, and E. Refused*. In this case, the LLM will output a probability for each of the tokens corresponding to the choices A through E, thereby generating a complete response distribution that we aim to align with the true distribution observed in survey data.

Formally, let $q \in Q$ be a question, $g \in G$ be a subpopulation, and $\mathcal{A}_q$ be the set of possible choices for question q. An LLM with parameters θ produces a conditional probability distribution $p_\theta(\mathcal{A}_q \mid q, g)$. We fine-tune this model so that its predicted distribution for each (q, g) mirrors the human response distribution $p_H(\mathcal{A}_q \mid q, g)$ collected from real survey data. To accomplish this, we apply LoRA fine-tuning (Hu et al., 2021) and use the forward Kullback–Leibler (KL) divergence as our loss. Concretely, if $p_H(\mathcal{A}_q \mid q, g)$ represents the group-level empirical distribution of human opinions and $p_\theta(\mathcal{A}_q \mid q, g)$ represents the model's predicted distribution, our training objective is:

$$\mathcal{L}(\theta) = \mathbb{E}_{q,g}\Big[D_{\mathrm{KL}}\big(p_H(\mathcal{A}_q \mid q,g) \,\big\|\, p_\theta(\mathcal{A}_q \mid q,g)\big)\Big],$$

where D_{KL} denotes the KL divergence. In the example shown in Figure 2, the model is trained to reduce the KL divergence between the target (survey-based) distribution over $\{A, B, C, D, E\}$ and its predicted distribution for the subpopulation living in the Southern United States.

We choose forward KL (i.e., $\mathrm{KL}(p_H \mid\mid p_\theta)$) since it is sensitive to cases where p_H assigns high probability but p_θ does not, naturally encouraging the model to *cover* the real distribution. This property aligns with standard maximum-likelihood training, where the model is penalized for underestimating any response that is frequent in the data. In other words, if many participants in group g choose option "A" for question q, then the model probability on "A" should be correspondingly high.

Instead of explicitly modeling the group response distribution as $p_H(\mathcal{A}_q|q, g)$, one could do two alternatives. (1) One-hot encoding: this approach (Li et al., 2024) approximates the distribution by a one-hot vector, assigning a value of one to the most probable option and zero elsewhere. (2) Data augmentation by response frequency: this approach (Zhao et al., 2023) expands the dataset by replicating question-choice pairs in proportion to their observed frequency. We adopt the explicit distribution modeling in our main experiments because it directly encodes the distributional information without requiring discrete sampling or replicating data points. A detailed comparison of these approaches is provided in Section C.1.

3.2 SubPOP: a Comprehensive Survey Dataset to Fine-tune and Evaluate LLMs

OpinionQA (Santurkar et al., 2023) is a widely used dataset for fine-tuning and evaluating large language models (LLMs) on opinion prediction, containing roughly 500 questions drawn from 14 American Trends Panel (ATP) waves (Pew Research Center, 2018). Although valuable, it faces two important limitations: (1) Limited thematic diversity—for instance, wave 26 focuses on the topic of firearms. (2) Reliance on a single survey family (ATP), which risks overfitting to a particular style of questions and limits out-of-distribution evaluation on other sources (e.g., GSS).

To address these limitations, we introduce a new dataset, SubPOP, that broadens both the thematic and institutional scope of opinion prediction data. For training, SubPOP comprises 3,229 multiple-choice questions drawn from ATP waves 61–132,

excluding waves included in OpinionQA. In Table 4, we list the topics of the ATP waves in SubPOP vs. OpinionQA, both showing the increased thematic diversity of SubPOP (with over 20 new topics) and the remaining unseen topics in OpinionQA that allow us to test whether LLMs fine-tuned on SubPOP can generalize to unseen topics.

For evaluation, SubPOP also includes 133 multiple-choice questions from the General Social Survey (GSS) (Davern et al., 2024), serving as an out-of-distribution benchmark. This expanded collection not only broadens the range of topics beyond OpinionQA's initial 500 questions, but also enables evaluation on surveys created and administered by different institutions (Pew Research Center vs. NORC-Chicago). Dataset curation and refinement pipeline is available in Appendix A.

3.3 Evaluation Metric

We use Wasserstein distance (WD) to quantify how closely the model's predicted opinion distribution matches human survey data (Santurkar et al., 2023; Moon et al., 2024; Meister et al., 2024; Zhao et al., 2023). Formally, for a group g representing some subpopulation and a question q WD is defined as $\mathcal{WD}_\theta(q,g) = \mathcal{WD}(p_H(\mathcal{A}_q|q,g), p_\theta(\mathcal{A}_q|q,g))$ (see formula in Appendix B). Since WD is computed over ordinal values, we map the categorical answer options to numbers, such as mapping "Very likely" to 1, "Likely" to 2, and so on.

Some prior work utilizes one-hot accuracy (Feng et al., 2024; Li et al., 2023) as an evaluation metric. However, one-hot accuracy only verifies whether the top-predicted choice matches the top human response, thereby discarding distributional information. In contrast, WD accounts for partial overlaps among the categories and reflects the 'cost' of shifting probability mass, providing a more nuanced assessment of distribution discrepancy. Consider the example question provided in Figure 2, where the human response distribution indicates that option B ("Somewhat likely") is the most probable. Now consider two cases in which the model incorrectly predicts the top choice. In the first case, the model assigns a high probability to option A ("Very likely"), while in the second case, it assigns a high probability to option D ("Very unlikely"). Although one-hot accuracy would treat both predictions equally as errors, WD differentiates between them by accounting for the ordinal relationship among the options, penalizing the second prediction more heavily for its larger deviation from the true distribution.

4 Experiments

4.1 Bounds of WD and Baselines

In this section, we describe the lower/upper bounds and two baseline methods against which we compare our method.

Lower and upper bounds. We use a uniform distribution over all available choices to establish an upper bound of the WD between a predicted and the target response distribution. To compute a lower bound, we sample a group of human respondents from the original human respondents to calculate the WD between the two, and perform bootstrapping to obtain a robust estimate. This lower bound captures the intrinsic variance arising from the respondent sampling process in opinion surveys.

Baselines. We compare our approach with two baseline methods: prompting and Modular Pluralism (Feng et al., 2024). For prompting, we consider both zero-shot and few-shot methods. In zero-shot prompting, we steer the LLM using demographic prompt formats. Specifically, we employ three different formats following Santurkar et al. (2023): QA, BIO, and PORTRAY. For instance, to condition the LLM to a person living in the South of the US, the QA format uses a question-answer format as illustrated in Figure 2; the BIO format conditions the model with a first-person narrative such as "I currently reside in the South."; and the PORTRAY format uses a third-person narrative like "Answer the following question as if you currently reside in the South.".

Few-shot prompting augments the prompt with a few examples of question-response distribution pairs alongside the demographic label (Hwang et al., 2023). In particular, we select the top five few-shot examples from the SubPOP training set based on cosine similarity computed by the embedding model. In our experiments, we represent the response distribution in JSON format and require the model to output its prediction in the same JSON format, following the approach in Meister et al. (2024).

Modular pluralism (Feng et al., 2024) fine-tunes multiple LLMs on distinct datasets to capture the viewpoints of different communities (Feng et al., 2023). For a given question, each fine-tuned LLM generates an opinion that reflects the perspective

Table 1: Evaluation on OpinionQA and the SubPOP evaluation set (SubPOP-Eval) for 22 subpopulations following (Santurkar et al., 2023). We compute the WD by averaging over all questions and subpopulations. Lower and upper bounds of performance give guidance on how each method performs. For Modular Pluralism, we provide an error rate of one-hot prediction (†) (Section 3.3) which was used in the original paper.

Method	OpinionQA				SubPOP-Eval			
	Llama-2-7B	Llama-2-13B	Mistral-7B	Llama-3-70B	Llama-2-7B	Llama-2-13B	Mistral-7B	Llama-3-70B
Upper bound (Unif.)	0.178				0.208			
Lower bound (Human)	0.031				0.033			
Zero-shot prompt (QA)	0.173	0.170	0.153	0.138	0.206	0.196	0.187	0.160
Zero-shot prompt (BIO)	0.193	0.183	0.162	0.143	0.221	0.212	0.202	0.175
Zero-shot prompt (PORTRAY)	0.195	0.207	0.158	0.209	0.212	0.242	0.194	0.247
Few-shot prompt	0.186	0.175	0.174	0.166	0.217	0.194	0.175	0.182
Modular Pluralism	0.285 (†55.6%)				0.279 (†55.2%)			
Ours (SubPOP-FT)	0.106	0.102	0.096	0.094	0.121	0.113	0.115	0.096

of the community it represents, and a separate black-box LLM aggregates these outputs to produce the final distributional response. Detailed implementation of the lower/upper bounds and the baselines is provided in Appendix D.

4.2 Generalization to Unseen Topics and Survey Families

In this section, we assess the ability of our fine-tuned LLMs to generalize to unseen data—both in terms of new topics and entirely different survey families. To evaluate these aspects, we use OpinionQA to measure generalization to unseen topics, and SubPOP-Eval to test generalization to a different survey family. We fine-tune four LLMs (Llama-2-7B, Llama-2-13B, Mistral-7B, and Llama-3-70B) on SubPOP-Train. We opt for pretrained LLMs rather than instruction-following models, as previous work has shown that pretrained models perform better on this task (Moon et al., 2024). A detailed comparison between these model types is provided in Appendix C.2.

Summary of results. Table 1 reports the average WD metrics computed over all demographic groups and survey questions, comparing our fine-tuned models against various baseline approaches. Our experiments show that fine-tuning on SubPOP-Train significantly outperforms all other methods, yielding a 32–46% reduction in WD on OpinionQA and a 39–42% reduction on SubPOP-Eval compared to the best baselines. Notably, SubPOP-Train is based on ATP data, while SubPOP-Eval is derived from GSS surveys—two distinct survey families that can differ in respondent pools, calibration techniques, and other methodological factors, leading to non-trivial distribution shifts despite both being representative of the US population. Furthermore, our fine-grained analyses at the wave level (see Appendix E) confirm that these trends persist even at more detailed levels of evaluation.

Comparison to zero- and few-shot prompting. We first compare the performance of prompting methods with our approach. Zero-shot prompting results in only modest WD improvements over the upper bound, with the largest gain observed for Llama-3-70B and negligible improvements for Llama-2-7B. Even when using few-shot prompting—where five example question-response distribution pairs are provided—the performance gains remain minimal. This may be partly due to an under-optimized prompt format (*e.g.* requiring JSON output) and the inherent sensitivity of language models to prompt formatting (Sclar et al., 2023; Anagnostidis and Bulian, 2024). These findings underscore the need for methods, such as fine-tuning, that enable relatively reliable predictions of opinion distributions.

Comparison to Modular Pluralism. Modular Pluralism improves one-hot accuracy, reducing prediction error from 72.7% (zero-shot prompting) to 55.6% on OpinionQA, but underperforms in matching the full distribution of option choices, measured as WD. This discrepancy in performance highlights the limitations of methods that train LLMs to identify only the most probable response rather than modeling the entire distribution of responses. Opinions are inherently distributed: even within a particular subpopulation such as a single demographic subgroup, distribution of opinions cannot be captured as a single most likely response. Moreover, instruction-tuned models that serve as a black-box LLM tend to assign high probabilities on only specific tokens (Lin et al., 2022; Kadavath et al., 2022; Achiam et al., 2023), further pushing the generated distribution away from the human distribution.